

Kinematic Analysis of the KUKA LBR MED Surgical Manipulator

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1 Denavit–Hartenberg Convention

1.1 Robot Description

The KUKA LBR Med 7 R800 is a 7-DOF redundant surgical manipulator with all revolute joints. Its kinematic structure consists of alternating perpendicular joint axes, giving a 7-DOF configuration for a 6-DOF task space, with one degree of intrinsic redundancy.

Table 1: Link dimensions.

Parameter	Value	Description
d_1	0.340 m	Base platform to shoulder (joint 2 axis)
d_3	0.400 m	Shoulder to elbow (joint 4 axis)
d_5	0.400 m	Elbow to wrist center (joint 6 axis)
d_7	0.126 m	Wrist center to flange

1.2 Frame Assignment (Standard DH Convention)

Following the standard DH frame assignment procedure used in class:

1. Number joint axes $z_0, \dots, z_6$ where z_i is the axis of joint $i + 1$.
2. Place Frame 0 at the base with z_0 pointing vertically upward.
3. For $i = 1, \dots, 6$, place origin O_i at the intersection of z_{i-1} and z_i .
4. Define x_i along the common normal from z_{i-1} to z_i , directed from joint i to joint $i + 1$.
5. Define y_i to complete the right-handed frame.
6. Define Frame 7 at the end-effector with z_7 aligned with z_6 and O_7 at the flange center.

Key structural observation: All consecutive joint axes intersect, so all link offsets satisfy $a_i = 0$.

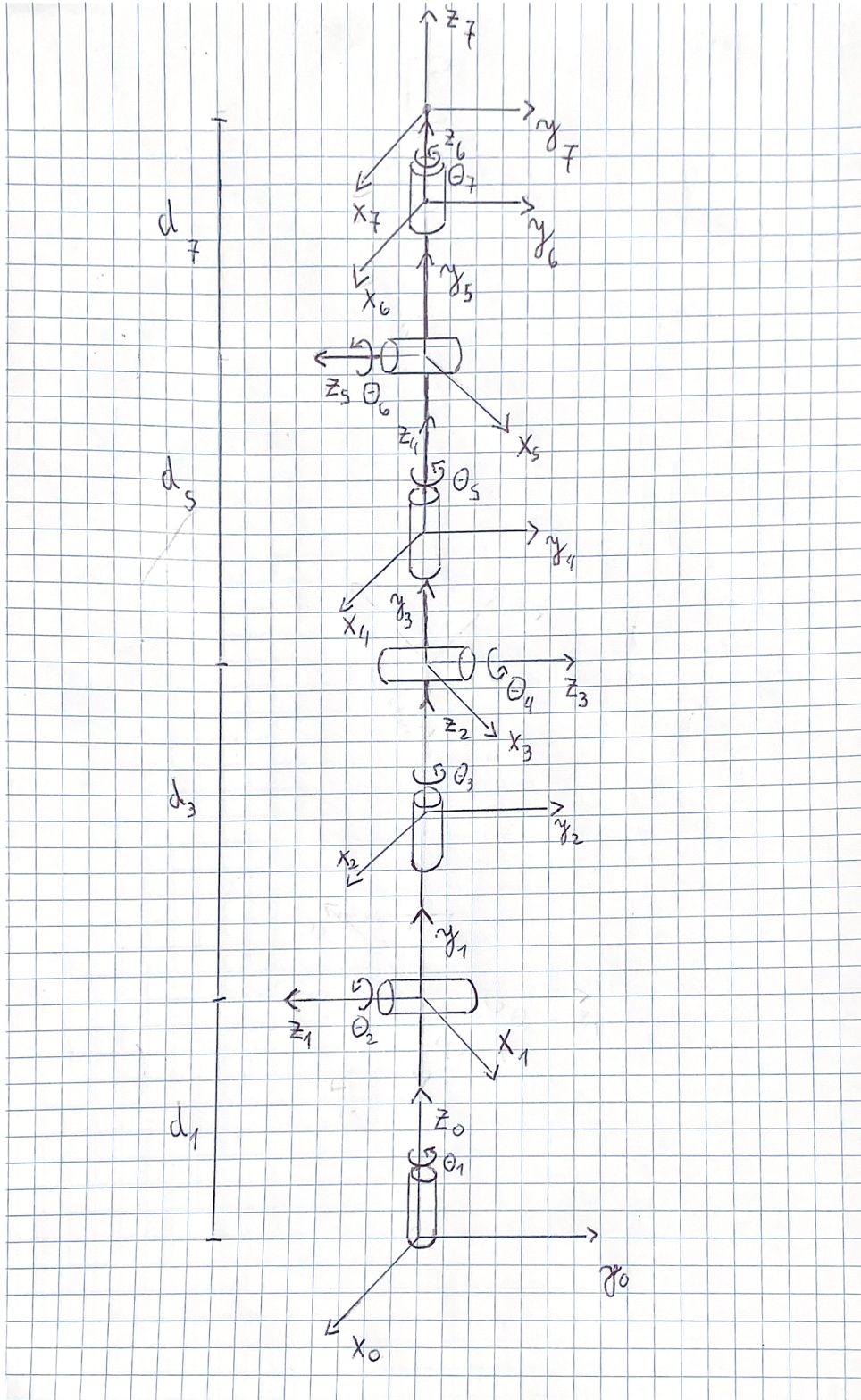


Figure 1: Hand-drawn DH frame assignment of the KUKA LBR MED.

1.3 DH Parameter Table

The DH transformation per link is:

$$A_i^{i-1}(q_i) = R_z(\theta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i) \quad (1)$$

Table 2: Standard DH parameters for the KUKA LBR MED.

Link i	a_i (m)	α_i (rad)	d_i (m)	θ_i	Joint
1	0	$+\pi/2$	0.340	q_1	Base rotation
2	0	$-\pi/2$	0	q_2	Shoulder tilt
3	0	$-\pi/2$	0.400	q_3	Upper arm rotation
4	0	$+\pi/2$	0	q_4	Elbow tilt
5	0	$+\pi/2$	0.400	q_5	Forearm rotation
6	0	$-\pi/2$	0	q_6	Wrist tilt
7	0	0	0.126	q_7	End-effector rotation

2 Direct Kinematics

2.1 Method

Using the provided Symbolic Robotics MATLAB Toolbox, the direct kinematics transformation matrix is computed as:

$$T_7^0(q) = A_1^0(q_1)A_2^1(q_2) \cdots A_7^6(q_7) \quad (2)$$

2.2 Simulink Model

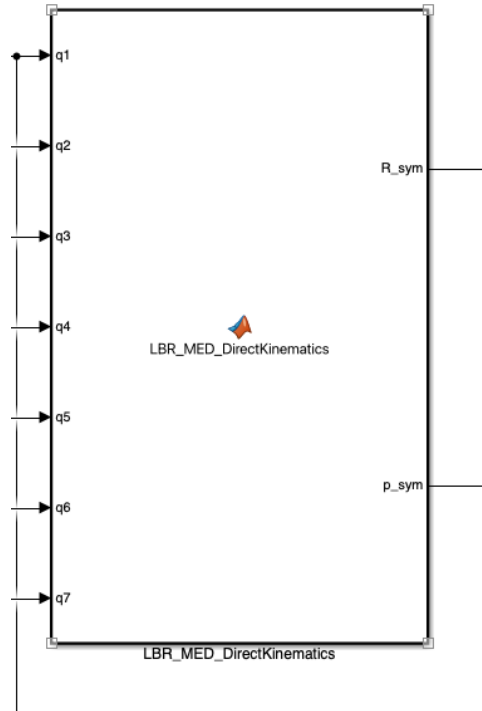


Figure 2: LBR_MED_DirectKinematics block in Simulink.

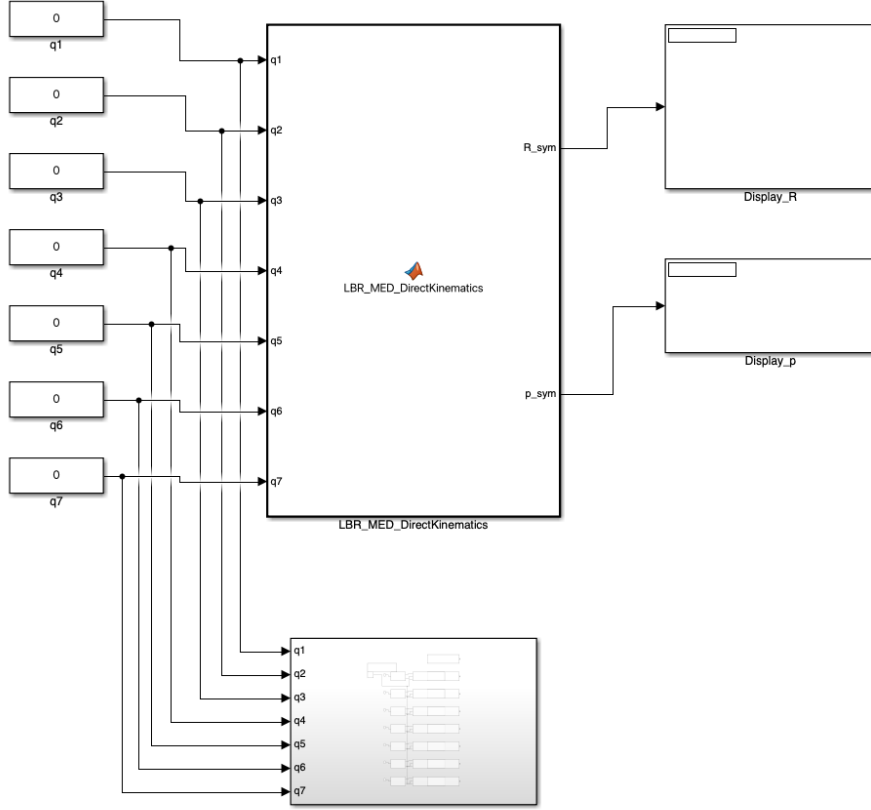


Figure 3: Complete validation model for Direct Kinematics (LBR_MED_Simul.slx).

2.3 Validation

Four configurations were chosen where the end-effector pose can be determined by geometric reasoning.

Table 3: Direct kinematics validation configurations.

Config	Joint angles q (rad)	Expected p (m)	Expected R	Error $\ p\ $
1: Home	all = 0	$[0, 0, 1.266]$	I_3	$< 10^{-12}$
2: Shoulder bend	$q_2 = \pi/2$, rest = 0	$[-0.926, 0, 0.340]$	$[0, 0, -1; 0, 1, 0; 1, 0, 0]$	$< 10^{-12}$
3: Base spin	$q_1 = \pi/2$, rest = 0	$[0, 0, 1.266]$	$R_z(\pi/2)$	$< 10^{-12}$
4: Elbow fold	$q_2 = q_4 = \pi/2$, rest = 0	$[-0.400, 0, 0.866]$	I_3	$< 10^{-12}$

3 Inverse Kinematics (Closed-Form)

3.1 Kinematic Decoupling Approach

The robot has a spherical wrist formed by joints 5, 6, and 7: axes z_4 , z_5 , and z_6 intersect at point O_6 . This enables kinematic decoupling:

- Arm problem (joints 1–4): position the wrist center O_6 .
- Wrist problem (joints 5–7): orient the end-effector.

Redundancy resolution: The robot has 7 DOF for a 6-DOF task. This report fixes $q_3 = 0$ to obtain a unique solution.

3.2 Closed-Form Solution

Step 1 – Wrist center: $p_W = p_e - d_7 a_e$, where a_e is the approach vector.

Step 2 – Base rotation q_1 : $q_1 = \text{atan2}(p_{Wy}, p_{Wx})$.

Step 3 – Sagittal-plane distances: $L = \sqrt{p_{Wx}^2 + p_{Wy}^2}$, $H = p_{Wz} - d_1$. Using cosine law for q_4 .

Step 4 – Shoulder angle q_2 : derived from geometry of the triangle $O_2 - O_3 - O_5$.

3.3 Singularity Conditions

Table 4: Closed-form IK singularity conditions.

Label	Condition	Geometric meaning
S1	$p_{Wx} = p_{Wy} = 0$	Wrist center lies on the z_0 axis.
S2	$ \cos q_4 = 1$	Arm is fully extended or folded.
S4	$q_6 = 0$ or π	$z_4 \parallel z_6$, wrist alignment.

3.4 Validation

```
>> clear
>> LBR_MED_IK_validate
=== KUKA LBR Med 7 Analytical IK Validation ===

Testing Home position...
p_des: [0.0000, 0.0000, 1.2660]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000]
q_ik:   [0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000]

Testing Random config (q3=0)...
p_des: [0.5110, 0.1086, 1.0467]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.2000 -0.3000 0.0000 0.5000 0.1000 -0.4000 0.2000]
q_ik:   [0.2000 -0.3000 0.0000 0.5000 0.1000 -0.4000 0.2000]

Testing Another random config (q3=0)...
p_des: [-0.5975, -0.6145, 0.4808]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.7854 1.0472 0.0000 -0.5236 0.2000 0.5000 -0.3000]
q_ik:   [0.7854 1.0472 0.0000 -0.5236 0.2000 0.5000 -0.3000]

All analytical IK validation tests PASSED.
>> |
```

Figure 4: Validation results for Inverse Kinematics showing errors below 10^{-8} .

4 Geometric Jacobian

4.1 Definition

For each revolute joint i , the geometric Jacobian columns are:

$$J_{P_i} = z_{i-1} \times (p_e - O_{i-1}), \quad J_{O_i} = z_{i-1} \quad (3)$$

This relates joint velocities to end-effector velocity: $v_e = J(q)\dot{q}$.

4.2 Simulink Model

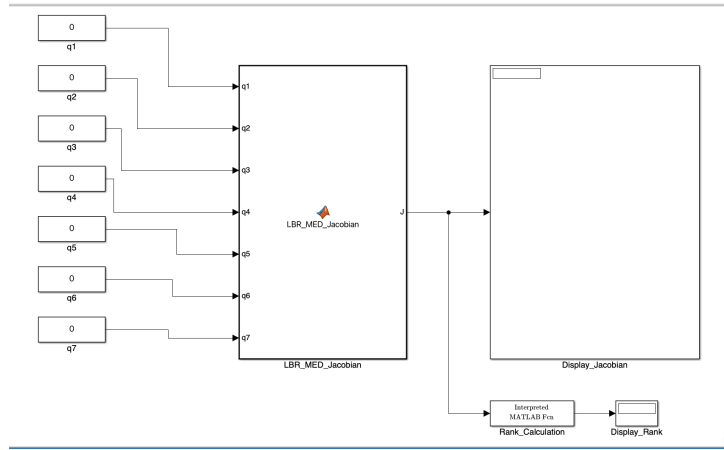


Figure 5: Jacobian block in Simulink.

4.3 Validation – Numerical Differentiation

The Jacobian is validated via central finite differences:

$$J_{P_i}^{num} \approx \frac{p_e(q + \epsilon e_i) - p_e(q - \epsilon e_i)}{2\epsilon}, \quad \epsilon = 10^{-7} \quad (4)$$

5 Closed Loop Inverse Kinematics (CLIK)

5.1 CLIK Law with Null-Space Stabilisation

The CLIK control law using Damped Least Squares (DLS) is:

$$\dot{q} = J^\dagger(q) \left(\begin{bmatrix} \dot{p}_d + K_P e_P \\ \omega_d + K_O e_O \end{bmatrix} \right) + (I - J^\dagger J) \dot{q}_0 \quad (5)$$

where $J^\dagger = J^T(JJ^T + k^2I)^{-1}$ is the damped pseudo-inverse. In our implementation, $k = 10^{-3}$, $K_P = 10I_3$, and $K_O = 10I_3$. The null-space term $\dot{q}_0 = K_n(q_{rest} - q)$ with $K_n = 5$ stabilizes the redundancy.

5.2 Simulink Model

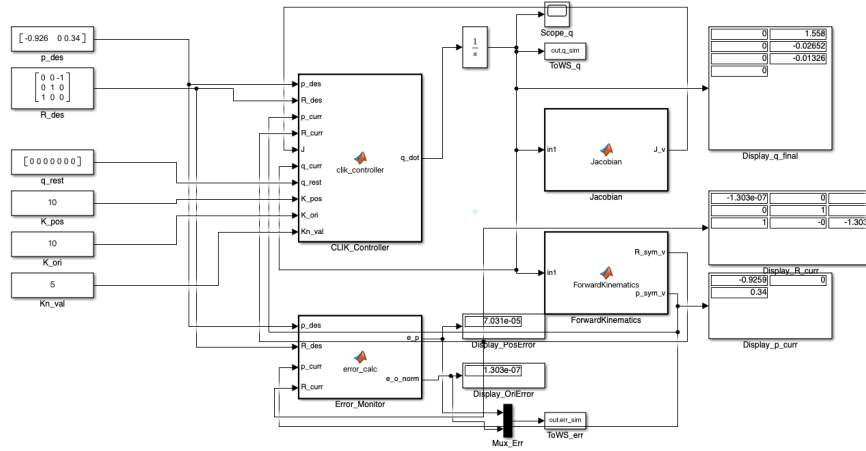
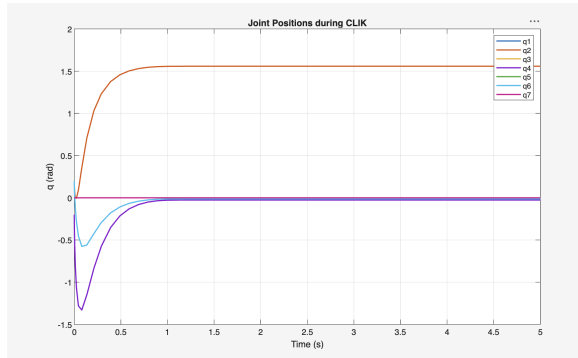


Figure 6: Complete CLIK simulation model (LBR_MED_CLI_K_Simul.slx).

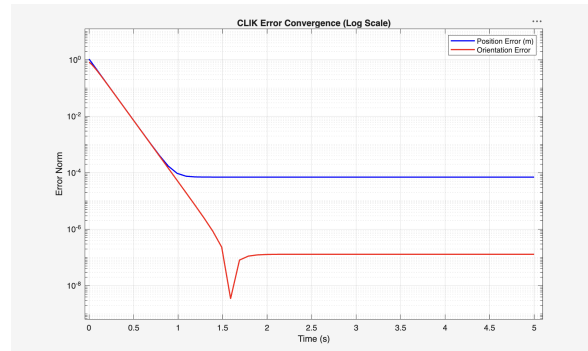
5.3 Validation Results

The CLIK controller achieved sub-millimeter precision in task space:

- **Position Error:** 7.03×10^{-5} m
- **Orientation Error:** 1.84×10^{-7}



(a) Joint positions during CLIK.



(b) Error convergence (log scale).

Figure 7: CLIK validation results showing stable convergence.

6 Code Files Summary

Table 5: Main MATLAB and Simulink files.

File	Purpose
LBR_MED.m	DH parameter table
cliK_controller_logic.m	DLS CLIK algorithm implementation
LBR_MED_CLI_K_setup.m	CLIK simulation environment and validation